

Unit 6, Lesson 8: Solving Real World Problems Involving Proportions

Benchmark

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

Learning Target

- ☐ I can solve problems using proportions.

6.8.1 Warm-Up: Proportions with Fractions

1. Place a digit in each blue box to make the equality statement true. Use only digits 1 to 9, at most one time each.

$$\frac{\overline{\boxed{}}}{\overline{\boxed{}}} = \frac{\overline{\boxed{}}}{\overline{\boxed{}}}$$

6.8.2 Exploration: Which Information Should I Use?

Elena is making large batches of fruit punch and marinated chicken for her upcoming family reunion.

She makes fruit punch by mixing 5 parts cranberry juice with 3 parts apple juice.

1. Complete the statement to explain what this means by writing numbers in each blank.

Regardless of the unit she uses to measure, Elena will use a ratio of cranberry to apple juice that is _____ to _____. For example, if she is measuring in cups, she would add _____ cups of apple juice for every _____ cups of cranberry juice she adds.

2. Elena plans to make 40 cups of fruit punch. Work with your partner to complete the proportion that could be used to find how much apple juice she needs for the fruit punch, a .

$$\frac{\quad}{\quad} = \frac{a}{40}$$

3. What does the numerator of each ratio represent in the proportion?
4. What does the denominator of each ratio represent in the proportion?
5. How did you use the values 5 and 3 to determine the numerator and denominator of the first ratio?
6. How much apple juice will Elena need to make 40 cups of fruit punch?

The chicken marinade recipe she uses says, "Mix 3 parts oil with 2 parts soy sauce and 1 part orange juice." Elena needs to make 13.5 cups of marinade for the amount of chicken she's marinating.

7. Create a proportion that could be used to find the amount of oil Elena needs for the recipe.
8. Elena is scaling the recipe up to make a larger batch. By what scale factor is Elena enlarging the recipe? Explain or show how you know.
9. How could this scale factor be used to determine how many cups of each ingredient she needs to make the marinade?
10. Complete the summary by writing numbers on each blank based on the situation.

If she's measuring in cups, Elena can make 6 cups of marinade using the original recipe. She is scaling the recipe up by a scale factor of _____ to make 13.5 cups of marinade. To do this, she will mix _____ cups of oil with _____ cups of soy sauce and _____ cups of orange juice.

6.8.3 Bring It Together: Proportions

Recall that ratios can be used to describe part-to-part relationships or part-to-whole relationships.

Situation: Elena’s fruit punch recipe uses 5 parts cranberry juice for every 3 parts apple juice.	
Part-to-Part Example	Part-to-Whole Example
The ratio of cranberry juice to apple juice is 5:3.	The ratio of cranberry juice to total fruit punch is 5:8.

When setting up proportions to solve problems, it’s important to first determine which quantities need to be compared to find the missing value.

1. Complete the table.

Situation: A teacher is planning a field trip to the aquarium. The aquarium requires 2 chaperones for every 15 students. The teacher plans accordingly and orders a total of 85 tickets.	
Set up a proportion that can be used to determine the number of students the teacher is planning to have on the field trip.	
Set up a proportion that can be used to determine the number of chaperones the teacher is planning to have on the field trip.	

2. Solve only one of the proportions above.
3. Explain how the solution to the proportion can be used to determine both the number of student tickets and the number of chaperone tickets the teacher ordered.

6.8.4 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar. How much oil is needed to make 19 teaspoons (tsp) of salad dressing?

2. A grocer can buy strawberries from a local farmer for \$1.38 per pound (lb). His last strawberry order cost \$241.50. How many pounds did the grocer order?

3. Andre and Han are moving boxes. Andre can move 4 boxes every half hour. Han can move 5 boxes every half hour. How long will it take Andre and Han to move all 72 boxes?

6.8.5 Wrap-Up: Ticket Out the Door

A recipe for sparkling grape juice calls for $1\frac{1}{2}$ quarts of sparkling water and $\frac{3}{4}$ quart of grape juice.

1. How much sparkling water should be mixed with 9 quarts of grape juice?
2. How much grape juice should be mixed with $\frac{15}{4}$ quarts of sparkling water?